

Big Data Processing

Big Data Processing Models

- Big data processing models are the programming models used to access large-scale data to extract useful information for supporting and providing decisions
- Big data processing use parallel programming models to improve the performance and speed up the computation process of the big data solutions.
- Big data processing Models can be classified into two categories: **Batch Processing Models** and **Real-time Processing Models**

Batch Data Processing

- Is a type of data processing where a batched accumulated data are collected, stored, and processed at a scheduled time.
- It is a type of data processing which accelerates the processing time and complete a series of tasks without additional human intervention.
- **MapReduce** is one of the most popular programming models for big data batch processing using large-scale commodity clusters. Map and Reduce functions are programmed to process the big data distributed across multiple heterogeneous nodes.
- **Hadoop** is the open-source big data framework which use **MapReduce** to perform big data batch process computation..

Big Data Processing

MapReduce Concept- Word Problem

map(key=url, val=contents):

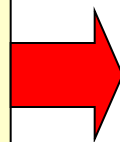
For each word w in contents, emit (w , "1")

reduce(key=word, values=uniq_counts):

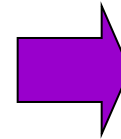
Sum all "1"s in values list

Emit result "(word, sum)"

see bob run
see spot throw



see	1
bob	1
run	1
see	1
spot	1
throw	1



bob	1
run	1
see	2
spot	1
throw	1

Big Data Processing

Real-time data processing

- Is an immediate type of processing where the data processing and updates occurs at the real time of event. It is the method where by data is received and processed at the same time through the use of internet functionalities.
- It captures data in a real-time and generate quick output updates
- Good examples of real-time data processing systems are bank ATMs, traffic control systems, etc
- Real-time data processing can be classified further into: **Transaction processing** and **Stream processing**.
- In the world of data analytics, stream processing is a common application of real-time data processing.

Stream Processing

- Stream data processing is a data processing of which a series of data processing is taken at the time of the data is created.
- It involves multiple tasks on the incoming series of data (the “data stream”), which can be performed serially, parallel, or both.
- This workflow is referred to as a stream processing pipeline, which includes the generation of the stream data, the processing of the data, and the delivery of the data to a final location.

Big Data Processing

Real-time data processing - Stream Processing

The following are some examples of Real-time data processing:

- **Real-time Sentiment Analysis:** Sentiment Analysis is a type of text analysis which systematically identify, extract, quantify, and study affective states and subjective information, For example to make a prediction for real-time sentiment analysis of Twitter feeds using Spark Streaming
- **Real-time Weather Data Analysis** – For example to make predictions on the occurrence of fog or haze in a specific location.

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Activity4: Individual Work (10 minutes)

Questions

1. Differentiate between Batch and Real-time data processing
2. List three software tools for
 - i. Batch data processing
 - ii. Real time data processing

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Differences between Batch Data Processing and Real-time data processing

	Batch Data Processing	Real-Time Data Processing
Hardware	Most storage and processing resources requirement to process large batches of data.	Less storage or no storage required to process the current data packets. Less computational requirements.
Performance	Latency could be minutes, hours, or days	Latency needs to be in seconds or milliseconds
Data set	Large batches of data	Few current data packet
Analysis	Complex computation and analysis of a larger time frame	Simple reporting or computation